Akoizo Original Practice - Form 1

Division C | Forensics | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which statement best expresses Locard exchange?

Unit: Forensic Method and Locard Exchange.
Worked-case data: A blue polyester fiber on a chair resembles fibers from a suspect's common jacket. The jacket was present in the room during an unrelated earlier meeting.
Separate investigation observations:
Material: A common blue polyester fiber is recovered.
Comparison: Several garments in the reference set share the same observed color and polymer class.
Context: One matching garment was in the room before the incident during ordinary use.

A. Every contact leaves a trace

B. Fingerprints are unique

C. DNA proves guilt alone

D. Confirmatory tests never err

Answer: _____

2. (4 points) Why run a distilled-water blank with Benedict reagent?

Unit: Lab Safety, QA, and Documentation.
Worked-case data: A reagent blank and an unknown both turn yellow. The expected negative blank should remain clear.
Separate investigation observations:
Positive control: The known target gives the expected response.
Reagent blank: The blank also gives a strong response that should not occur.
Unknown: The unknown produces the same response.

Answer: _____
_____
_____

3. (2 points) Powder fizzes with HCl and gives basic pH. Best inference?

Unit: Powder Scheme: Solubility, pH, and Precipitation.
Worked-case data: The exercise candidate set is sodium chloride, sodium carbonate, and calcium carbonate. The unknown reacts with acid to release gas and is poorly soluble in water.
Separate investigation observations:
Water solubility: Most of the white solid remains undissolved.
Acid response: A fresh aliquot releases gas; the blank remains negative.
Reference comparison: In this candidate set, calcium carbonate is poorly soluble and acid-reactive; sodium carbonate is soluble; sodium chloride is not acid-effervescent.

A. Sucrose

B. Starch

C. Chloride salt

D. Carbonate

Answer: _____

4. (2 points) Every sample and the tool blank show yellow emission. Best next step?

A. Mix all unknowns

B. Accept every sample as pure sodium salt

C. Resolve carryover and obtain a clean blank

D. Ignore the blank

Answer: _____

5. (2 points) Which polymer is a condensation product?

Unit: Polymer Chemistry and Recycling Codes.
Worked-case data: A fragment floats in water. The reference set contains polypropylene and two polyethylene samples with densities below water.

A. HDPE

B. PET

C. PP

D. PS

Answer: _____

6. (2 points) Green Beilstein flash means:

Unit: Plastic ID: Density, Flame, and Beilstein.
Worked-case data: A fragment sinks in a 0.92 g/mL reference liquid and floats in a 0.98 g/mL reference liquid. A candidate table lists 0.90, 0.95, and 1.20 g/mL samples.

A. Halogen such as PVC chlorine

B. Starch

C. Glucose

D. Sodium

Answer: _____

7. (2 points) Twisted ribbon, paper odor, soft ash. Fiber?

Unit: Fibers: Classification and Microscopy.
Worked-case data: The questioned fiber is a flattened twisted ribbon with a cellulose-associated reference response. Reference A is cotton, B is wool, and C is a uniform synthetic filament.
Separate investigation observations:
Longitudinal view: A flattened ribbon with repeated twists is visible.
Composition record: The supplied reference assay supports a cellulose-based material.
Comparison set: Wool shows scales; the synthetic reference is a smooth uniform filament; cotton shows convolutions.

A. Nylon

B. Polyester

C. Wool

D. Cotton

Answer: _____

8. (4 points) Medulla width 40 um, shaft 60 um. Index and inference?

Answer: _____
_____
_____

9. (4 points) Spot 4.2 cm, front 7.0 cm. Rf?

Answer: _____
_____
_____

10. (2 points) Tallest peak in a mass spectrum is called:

Unit: Mass Spectrometry and Spectroscopy.
Worked-case data: A spectrum contains a strong m/z 91 fragment. Candidate A and candidate B can both produce that fragment, but their molecular masses differ.
Separate investigation observations:
Base peak: The largest signal is m/z 91.
Candidate comparison: Two candidates in the supplied reference set produce m/z 91.
Mass constraint: The intact-ion region is too weak to interpret reliably in this record.

A. Base peak

B. M+2

C. Solvent front

D. Molecular ion

Answer: _____

11. (2 points) Loop vs whorl delta count?

Unit: Prints: Patterns and Minutiae.
Worked-case data: A questioned print and a reference both show a loop. The questioned image has a large smear through the region where detailed features would be compared.
Separate investigation observations:
Ridge flow: Both images show a loop pattern.
Detail quality: Most of the questioned print is smeared; only two short clear segments remain.
Candidate context: Several reference prints share the same pattern class.

A. Both 0

B. Both 2

C. Loop 1, whorl 2

D. Loop 2, whorl 1

Answer: _____

12. (2 points) What should guide method selection for a paper note?

Unit: Developing Latent Prints.
Worked-case data: Exhibit A is an uncoated paper note. Exhibit B is a smooth plastic container. Both may have latent residues.
Separate investigation observations:
Surface: The exhibit is uncoated absorbent paper.
Residue: The question concerns amino-acid-containing latent residue.
Other evidence: The note also has ink that may require separate comparison.

A. The darkest reagent

B. Porosity, residue chemistry, and other evidence needs

C. A universal method order for every surface

D. The suspect's identity

Answer: _____

13. (2 points) Heterozygote at one STR locus shows:

Unit: DNA: STR Profiling and Electrophoresis.
Worked-case data: The questioned profile is locus 1: 10,12 and locus 2: 8,9. Reference A is 10,12 and 8,9. Reference B is 10,11 and 8,9.
Separate investigation observations:
Questioned profile: Locus 1: 10,12. Locus 2: 8,9.
Reference profiles: A: 10,12 / 8,9. B: 10,11 / 8,9.
Quality statement: This teaching dataset is complete, uncontaminated, and single-source; no dropout is modeled.

A. One peak

B. Two peaks

C. No peaks

D. Three peaks

Answer: _____

14. (2 points) Parents AO x BO can have child type:

Unit: Blood: ABO Genetics and Spatter.
Worked-case data: A stain has a 6 mm minor axis and a 12 mm major axis. Its valid typing reacts only with anti-A.

A. A, B, AB, or O

B. AB only

C. B only

D. O only

Answer: _____

15. (2 points) Radial cracks make right angles on which side?

Unit: Glass Fracture and Ballistics.
Worked-case data: A questioned fragment measures RI 1.520 ± 0.002. Reference A is 1.521 ± 0.002; reference B is 1.535 ± 0.002 under comparable conditions.
Separate investigation observations:
Questioned fragment: RI 1.520 ± 0.002.
Reference A: RI 1.521 ± 0.002 under the same stated conditions.
Reference B: RI 1.535 ± 0.002 under the same stated conditions.

A. Both equally

B. Neither

C. Impact side

D. Reverse from impact

Answer: _____

16. (4 points) 24 h at 22 C with base 6 C gives ADH?

Answer: _____
_____
_____

17. (4 points) Rewrite as CER: powder is CaCO3.

Answer: _____
_____
_____

18. (4 points) Why can one reliable exclusion outweigh several class matches?

Unit: Mock Crime Synthesis and Time Plan.
Worked-case data: Candidate A's pen is compatible with a questioned ink pattern. A complete single-source stain profile excludes A as its donor. The scenario does not state that the writer and stain donor must be the same person.
Separate investigation observations:
Ink comparison: A's pen and the questioned ink have compatible developed patterns.
DNA comparison: A is excluded as sole donor of a separate clean single-source stain.
Scenario constraint: The case does not require the writer and stain donor to be the same person.

Answer: _____
_____
_____